

Literature-Implied Answer To Problem 4.1 Of arXiv:2107.04662

Run fa.banach_001

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Status

literature_implied_answer (full problem). This packet records that Problem 4.1 from Kakol–Leiderman [1] is answered negatively by combining the problem statement with a no-surjection theorem recalled in Kakol–Kurka–Sliwa [2], attributed there to Leiderman–Levin–Pestov. The relation is an agent-identified implication; the supporting source does not explicitly say that it is resolving Problem 4.1.

Source Question

Kakol–Leiderman ask, in Problem 4.1 on PDF page 10:

Problem 4.1. *Does there exist an infinite compact space X admitting a continuous linear surjection $T : C_p(X) \rightarrow C_k(X)_w$?*

In Theorem [4.4](#) we show that such X (if exists) cannot be an infinite metrizable compact C -space. In particular, X cannot be an infinite metrizable finite-dimensional compact space, equivalently, an infinite compact subspace of the Euclidean space $\mathbb{R}^n$.

In text, the question is whether there exists an infinite compact space X admitting a continuous linear surjection

$$T : C_p(X) \longrightarrow C_k(X)_w.$$

Here $C_k(X)_w$ denotes the Banach space $C(X)$, with its compact-open norm topology forgotten and replaced by the weak topology.

Supporting Theorem

Kakol–Kurka–Sliwa [2], in the introduction immediately after their Problem 1, recall the following restriction from the earlier work of Leiderman–Levin–Pestov:

If X is a Tychonoff space and E is an infinite-dimensional normed space, then there exists no sequentially continuous linear surjection from $C_p(X)$ onto E_w .

Identification

Assume that X is an infinite compact Hausdorff space and that

$$T : C_p(X) \longrightarrow C_k(X)_w$$

is a continuous linear surjection. Put $E = C(X)$, regarded as a normed space with the usual supremum norm. Since X is infinite compact Hausdorff, $C(X)$ is infinite-dimensional: for every finite set of distinct points of X , normality gives continuous functions separating those points, so $C(X)$ has arbitrarily large finite-dimensional subspaces.

The codomain $C_k(X)_w$ is precisely E_w . Moreover, every continuous map is sequentially continuous. Thus T is a sequentially continuous linear surjection from $C_p(X)$ onto E_w , contradicting the quoted Leiderman–Levin–Pestov restriction. Hence no such infinite compact space X exists.

Scope And Novelty Check

This is not claimed as a new theorem of the run. It is a full negative answer to the extracted Problem 4.1 by direct identification with a known no-surjection theorem. The source paper itself proves nonexistence for infinite metrizable compact C -spaces; the literature implication above rules out all infinite compact X .

Cheap run indexes were searched for ‘2107.04662’, ‘C_k(X)_w’, ‘linear surjection’, and related terms, with no exact prior packet found. Local source search found the later arXiv paper ‘2509.08634’ quoting the decisive restriction, and web search found the same two primary arXiv sources.

References

- [1] J. Kakol and A. Leiderman, *On linear continuous operators between distinguished spaces $C_p(X)$* , arXiv:2107.04662, 2021.
- [2] J. Kakol, O. Kurka, and W. Sliwa, *Lotz-Peck-Porta and Rosenthal’s theorems for spaces $C_p(X)$* , arXiv:2509.08634, 2025.